Entropy of the Hidden-Manifold Output Distribution

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Abstract

We compute the quenched entropy of the output distribution of a hidden-manifold model with a generic channel P_{out}, using the replica method. The full derivation is presented down to the replica-symmetric saddle-point equations.

Contents

1 Introduction

1.1 TODO (Agents should not edit this section)

- Outer-1RSB ansatz set up in Section APPENDIX E (breaks the disorder index α, keeps the $(n+1)$-block RS); explicit replicon/stability computation around the $q_0 = 0$ RS saddle for the sign channel still pending.
- Add Bayesian interpretation of overlaps.
- Add countinuous activations case, e.g. erf(h) + noise
- Compute Fisher information?

1.2 The hidden manifold model

The **hidden manifold model** (HMM) [1] is a generative framework for high-dimensional structured data that plays a central role in the statistical-physics theory of supervised learning. Observations $x \in \mathbb{R}^N$ are produced by projecting a low-dimensional Gaussian latent code $z \in \mathbb{R}^D$ through a random linear map $F \in \mathbb{R}^{N \times D}$, followed by an entry-wise output channel P_{out}:

$$z \sim \mathcal{N}(0, I_D), \quad x_i \sim P_{\text{out}}(\cdot \mid (Fz)_i), \quad F_{i\mu} \sim \mathcal{N}\left(0, \frac{1}{D}\right) \quad \text{i.i.d.} \tag{1}$$

The aspect ratio $\gamma = \frac{N}{D}$ interpolates between the under-determined ($\gamma < 1$) and over-determined ($\gamma > 1$) regimes and controls the richness of the output distribution $p_F(x)$.

The HMM captures the empirical observation that real data concentrates near a lower-dimensional manifold, while retaining Gaussian-equivalence properties that make the replica method analytically tractable [1]. It has become a standard framework for deriving exact asymptotics of generalization in the proportional limit $N, D \to \infty$ at fixed γ. [2] used the replica method to compute the exact generalization error for ridge regression and random features under the HMM, establishing a rigorous Gaussian equivalence theorem that reduces the non-linear model to an effective Gaussian problem. [3] extended this analysis to generic feature maps and fully characterized the learning-curve phenomenology, including the double-descent peak and the under-to-over-parametrized crossover.

1.3 Setup and goals

These notes address a complementary question: rather than studying how well a learner can recover a target function, we study the **information content of the output distribution $p_F(x)$ itself**. Let

$$p_F(x) = \int Dz \prod_{i=1}^{N} P_{\text{out}}(x_i \mid (Fz)_i), \quad Dz = (2\pi)^{-\frac{D}{2}} e^{-|z|^2_{\frac{}{2}}}\,\mathrm{d}z, \tag{2}$$

with $F_{i\mu} \sim \mathcal{N}\left(0, \frac{1}{D}\right)$, quenched, and $N, D \to \infty$ at fixed

$$\gamma = \frac{N}{D}. \tag{3}$$

We compute the **quenched entropy**

$$S = \mathbb{E}_F[H_F(X)], \quad H_F(X) = -\int \mathrm{d}x\, p_F(x) \log p_F(x), \tag{4}$$

and its Rényi generalization $S_\alpha = \mathbb{E}_F[H_\alpha(X)]$ for $\alpha > 0$, using the replica method.

1.4 Outline

The derivation proceeds as follows.

Section 2 reduces the Shannon entropy to a replicated partition function, introduces the overlap order parameters via Hubbard–Stratonovich, and derives the replica-symmetric (RS) saddle-point equations. A key structural feature is the Bayes-optimal Nishimori collapse $q_0^* = 0$, $q_1^* = q^*$, with conjugates $\hat{q}_0^* = \hat{q}_d^* = 0$ and $\hat{q}_1^* = \hat{q}^*$, which reduces the five RS parameters to a single overlap q^* satisfying a Fisher-information self-consistency equation. A dedicated subsection clarifies what is exact and what is an ansatz in the **outer** (disorder-replica, m) space: at $n = 0$ the identity $Z_0(F) = 1$ makes outer RS automatic at the order needed for the Shannon entropy, whereas the inner-block RS assumption is a Bayes-optimal ansatz justified by Nishimori exchangeability and the standard local-stability hypothesis. At finite Rényi order the situation reverses — RS is genuinely an ansatz, parity symmetry forces $q_0 = 0$ on the RS branch but does not exclude outer 1RSB, and a replicon check is needed.

Section 3 extends the machinery to the quenched Rényi entropy S_α at finite order $\alpha = n + 1$. First steps toward an 1RSB computation are done in Section APPENDIX E. **Section 4** works out the noiseless sign channel in detail, recovering the Cover dichotomy Equation 87 at $\alpha \to 0$, providing closed or semi-closed expressions for the Shannon, collision, and min-entropy, and reducing the saddle-point system to a single scalar equation under the empirically observed $q_0 = 0$ ansatz; Figure 1 gives the full numerical Rényi profile $s_{\alpha(\gamma)}$ across $\alpha \in [0.02, 10]$.

Heavier algebraic details are collected in the appendices.

2 Replica computation of the entropy

2.1 Replica representation of the entropy

2.1.1 Replicas of the entropy at fixed F

Since $\int \mathrm{d}x\, p_F(x) = 1$, the identity

$$-\int \mathrm{d}x\, p_F(x) \log p_F(x) = -\partial_n \int \mathrm{d}x\, p_F(x)^{n+1} \Big|_{n=0} \tag{5}$$

holds. Define, for integer $n \geq 0$,

$$Z_n(F) := \int \mathrm{d}x\, p_F(x)^{n+1}. \tag{6}$$

Because $Z_0(F) = 1$, we have $\log Z_0(F) = 0$, and so

$$H_F(X) = -\partial_n \log Z_n(F)\mid_{n=0}. \tag{7}$$

Expanding the $(n+1)$-th power introduces $n+1$ latent replicas $z^a \in \mathbb{R}^D$, $a = 0, ..., n$:

$$Z_n(F) = \int \mathrm{d}x \prod_{a=0}^{n} Dz^a \prod_{i=1}^{N} \prod_{a=0}^{n} P_{\text{out}}\big(x_i \mid (Fz^a)_i\big). \tag{8}$$

The same observation x_i is shared by all replicas: this coupling is the structural source of the overlap order parameters.

2.1.2 Quenched average: second replica index

We need $S = -\mathbb{E}_F \partial_n \log Z_n(F)|_{n=0}$. Introduce a disorder-replica index $\alpha = 1, ..., m$:

$$\mathbb{E}_F \log Z_n(F) = \lim_{m \to 0} \frac{1}{m} \log \mathbb{E}_F[Z_n(F)^m], \tag{9}$$

so that

$$S = -\partial_n \lim_{m \to 0} \frac{1}{m} \log \mathbb{E}_F[Z_n(F)^m] \Bigg|_{n=0}. \tag{10}$$

For integer m,

$$Z_n(F)^m = \int \prod_{\alpha=1}^{m} \mathrm{d}x^\alpha \prod_{\alpha,a} Dz^{\alpha a} \prod_{i=1}^{N} \prod_{\alpha,a} P_{\text{out}}\big(x_i^\alpha \mid (Fz^{\alpha a})_i\big). \tag{11}$$

We collect the replica indices into $r = (\alpha, a)$, $r = 1, ..., M$, $M = m(n+1)$.

2.2 Gaussian disorder average and order parameter

2.2.1 Gaussian average over F

The disorder F enters only through the linear forms $h_i^r := (Fz^r)_i$. For fixed z's, (h_i^r) is jointly Gaussian with covariance

$$\mathbb{E}_F\left[h_i^r h_j^s\right] = \delta_{ij} Q_{rs}, \quad Q_{rs} := \frac{1}{D} z^r \cdot z^s, \tag{12}$$

and is i.i.d. across the N sites. Thus

$$\mathbb{E}_F[Z_n(F)^m] = \int \prod_r Dz^r \,[Z_{\text{out}}(Q)]^N, \tag{13}$$

$$Z_{\text{out}}(Q) := \int \prod_\alpha \mathrm{d}x^\alpha \int \mathrm{d}\mu_Q(h) \prod_{\alpha,a} P_{\text{out}}(x^\alpha \mid h^{\alpha a}), \tag{14}$$

where $\mathrm{d}\mu_Q(h)$ is the centered M-dim. Gaussian measure with covariance Q.

2.2.2 Hubbard–Stratonovich and the prior action

The integrand of Equation 13 depends on z only via the symmetric matrix $Q_{rs} = z^r \cdot z^s / D$. Inserting a delta-function and Fourier representation introduces the conjugate $\hat{Q}$, after which the z-integral factorizes over the D components of z^r and is Gaussian:

$$\int \prod_r Dz^r \exp\left(\frac{1}{2} \sum_{rs} \hat{Q}_{rs}\, z^r \cdot z^s\right) = \left[\det\big(I - \hat{Q}\big)\right]^{-\frac{D}{2}}. \tag{15}$$

Collecting all D-extensive contributions,

$$\mathbb{E}_F[Z_n(F)^m] = \int \mathrm{d}Q\, \mathrm{d}\hat{Q} \exp\Big\{D\, \mathcal{S}_{m,n}\Big(Q, \hat{Q}\Big)\Big\}, \tag{16}$$

$$\mathcal{S}_{m,n}\Big(Q, \hat{Q}\Big) = \mathcal{S}_{\text{prior}}\Big(Q, \hat{Q}\Big) + \gamma\, \mathcal{S}_{\text{out}}(Q), \tag{17}$$

$$\mathcal{S}_{\text{prior}} = -\frac{1}{2} \operatorname{Tr}\Big(Q\hat{Q}\Big) - \frac{1}{2} \log\det\Big(I - \hat{Q}\Big), \quad \mathcal{S}_{\text{out}} = \log Z_{\text{out}}(Q). \tag{18}$$

In the limit $D \to \infty$, saddle-point evaluation gives $\frac{1}{D} \log \mathbb{E}_F[Z_n(F)^m] \to \operatorname{extr}_{Q,\hat{Q}} \mathcal{S}_{m,n}$, and the entropy density is

$$s := \lim_{N\to\infty} \frac{S}{N} = -\frac{1}{\gamma}\, \partial_n \lim_{m\to 0} \frac{1}{m} \operatorname*{extr}_{Q,\hat{Q}} \mathcal{S}_{m,n} \bigg|_{n=0}. \tag{19}$$

2.3 The replica-symmetric calculation

2.3.1 The RS ansatz

We impose invariance under permutations of the disorder replicas α and, separately, under permutations of the entropy replicas a within each block, giving

$$Q_{(\alpha,a),(\beta,b)} = \begin{cases} 1 & \alpha = \beta, a = b \\ q_1 & \alpha = \beta, a \neq b \\ q_0 & \alpha \neq \beta \end{cases} \tag{20}$$

$$\hat{Q}_{(\alpha,a),(\beta,b)} = \begin{cases} \hat{q}_d & \alpha = \beta, a = b \\ \hat{q}_1 & \alpha = \beta, a \neq b \\ \hat{q}_0 & \alpha \neq \beta. \end{cases} \tag{21}$$

The diagonal of Q is 1 because the Gaussian prior fixes $|z^r|^2/D \to 1$.

The matrix $\hat{Q}$ has three eigenvalues coming from invariant subspaces (within-block antisymmetric, between-block antisymmetric, fully symmetric); see Section APPENDIX A for the diagonalization. The result is

$$\log\det\Big(I - \hat{Q}\Big) = mn \log A + (m-1) \log B + \log C, \tag{22}$$

with

$$\begin{aligned} A &= 1 - \hat{q}_d + \hat{q}_1, \\ B &= 1 - \hat{q}_d - n\hat{q}_1 + (n+1)\hat{q}_0, \\ C &= 1 - \hat{q}_d - n\hat{q}_1 - (m-1)(n+1)\hat{q}_0. \end{aligned} \tag{23}$$

The trace term enumerates immediately to

$$\operatorname{Tr}\Big(Q\hat{Q}\Big) = m(n+1)\hat{q}_d + m\, n(n+1)\, q_1\hat{q}_1 + m(m-1)(n+1)^2\, q_0\hat{q}_0. \tag{24}$$

2.3.2 Output action under the RS ansatz

The RS structure of Q admits the hierarchical Gaussian decomposition

$$h^{\alpha a} = \sqrt{q_0}\, u + \sqrt{q_1 - q_0}\, v^\alpha + \sqrt{1 - q_1}\, w^{\alpha a}, \tag{25}$$

with $u, \{v^\alpha\}, \{w^{\alpha a}\}$ mutually independent standard Gaussians; the covariance reproduces $1, q_1, q_0$ on the three RS strata. Define

$$\tilde{\mathcal{P}}(x \mid z) := \int Dw\, P_{\text{out}}\Big(x \;\Big|\; z + \sqrt{1 - q_1} w\Big), \tag{26}$$

$$\mathcal{P}(x \mid u, v) := \tilde{\mathcal{P}}\big(x \;\big|\; \sqrt{q_0} u + \sqrt{q_1 - q_0} v\big). \tag{27}$$

Both are normalized in x. Because the $w^{\alpha a}$ are independent across a, the a-product factorizes into $[\mathcal{P}(x^\alpha \mid u, v^\alpha)]^{n+1}$, and the disorder replicas decouple as well. We obtain

$$Z_{\text{out}}(q_0, q_1) = \int Du\,[\mathcal{K}_n(u; q_0, q_1)]^m, \quad \mathcal{K}_n := \int Dv \int \mathrm{d}x\,[\mathcal{P}(x \mid u, v)]^{n+1}. \tag{28}$$

2.3.3 Limit $m \to 0$

Since $\mathcal{K}_0 = 1$, expanding $\exp(m \log \mathcal{K}_n)$ gives

$$\frac{1}{m} \mathcal{S}^{\text{RS}}_{\text{out}} |_{m\to 0} = \int Du \log \mathcal{K}_n(u; q_0, q_1) =: \Phi_n(q_0, q_1). \tag{29}$$

For the prior, expanding Equation 22 for small m (using $C - B = -m(n+1)\hat{q}_0$, see Section APPENDIX B) gives

$$\frac{1}{m} \log\det\big(I - \hat{Q}\big) |_{m\to 0} = n \log A + \log B_0 - \frac{(n+1)\hat{q}_0}{B_0}, \tag{30}$$

$$B_0 := 1 - \hat{q}_d - n\hat{q}_1 + (n+1)\hat{q}_0, \tag{31}$$

and the trace term divides cleanly by m. Combining,

$$\begin{aligned} \frac{1}{m} \mathcal{S}^{\text{RS}}_{\text{prior}} |_{m\to 0} = &-\frac{1}{2}[(n+1)\hat{q}_d + n(n+1)q_1\hat{q}_1 - (n+1)^2 q_0\hat{q}_0] \\ &-\frac{1}{2}\left[n \log A + \log B_0 - \frac{(n+1)\hat{q}_0}{B_0}\right]. \end{aligned} \tag{32}$$

Set

$$\varphi_n := \lim_{m\to 0} \frac{1}{m} \mathcal{S}^{\text{RS}}_{m,n} = \big(\mathcal{S}^{\text{RS}}_{\text{prior}}/m\big)|_{m\to 0} + \gamma\,\Phi_n(q_0, q_1), \tag{33}$$

so that

$$s = -\frac{1}{\gamma}\,\partial_n \operatorname{extr} \varphi_n |_{n=0}. \tag{34}$$

2.4 The Shannon limit $n \to 0$

2.4.1 Output term: conditional output entropy

In Equation 27, $\mathcal{P}$ depends on (u, v) only through $z := \sqrt{q_0}u + \sqrt{q_1 - q_0}v$. With u, v standard normal, $z \sim \mathcal{N}(0, q_1)$. Hence

$$\mathcal{K}_n(u) = \int Dv \int \mathrm{d}x\,\Big[\tilde{\mathcal{P}}(x \mid z(u,v))\Big]^{n+1}. \tag{35}$$

At $n = 0$, the x-integral gives 1, so $\mathcal{K}_0 = 1$ and $\Phi_0 = 0$. Differentiating once,

$$\partial_n \int \mathrm{d}x\,\Big[\tilde{\mathcal{P}}(x|z)\Big]^{n+1} |_{n=0} = -H\big(\tilde{\mathcal{P}}(\cdot\,|z)\big), \tag{36}$$

so

$$\partial_n \mathcal{K}_n(u)|_{n=0} = -\mathbb{E}_v\Big[H\big(\tilde{\mathcal{P}}(\cdot\,|z(u,v))\big)\Big] \tag{37}$$

and

$$\boxed{\partial_n \Phi_n |_{n=0} = -\mathbb{E}_{z\sim\mathcal{N}(0,q_1)}\Big[H\big(\tilde{\mathcal{P}}(\cdot \mid z)\big)\Big].} \tag{38}$$

This is the conditional output entropy at common-cause variance q_1.

2.4.2 Prior term

Differentiating Equation 32 in n at fixed parameters and evaluating at $n = 0$ (envelope theorem at the saddle), one obtains, with $B := 1 - \hat{q}_d + \hat{q}_0$,

$$\partial_n \left(\frac{1}{m}\mathcal{S}^{\mathrm{RS}}_{\mathrm{prior}}\right)\,|_{n=0} = -\frac{1}{2}[\hat{q}_d + q_1\hat{q}_1 - 2q_0\hat{q}_0] - \frac{1}{2}\left[\log A - \frac{\hat{q}_1}{B} + \frac{\hat{q}_0(\hat{q}_0 - \hat{q}_1)}{B^2}\right]. \tag{39}$$

The detailed expansion is given in Section APPENDIX C.

2.5 Saddle-point equations and final formula

2.5.1 RS saddle equations and the Bayes-optimal collapse

The bare n^0 stationarity equations of φ_n are degenerate (since $\log Z_0 = 0$ identically forces $\varphi_0 = 0$ at the saddle). The physical equations come from the order-n piece. Carrying out the standard expansion (Section APPENDIX D), one finds for the Bayes-optimal entropy of the hidden-manifold model the Nishimori-type collapse

$$q_0^* = 0, \quad q_1^* = q^*, \tag{40}$$

leaving a single scalar overlap $q^* \in [0, 1]$.

The collapse has a transparent meaning once one identifies the role each replica plays in the Bayes-optimal setup. The marginal $p_F(x)$ is the law of x in the **teacher-student** generative model

$$z^* \sim \mathcal{N}(0, I_D), \quad x_i \sim P_{\mathrm{out}}\big(\cdot \mid (Fz^*)_i\big), \tag{41}$$

with the same Gaussian prior on z^* as in Equation 6 — sampling x from p_F is equivalent to first drawing a teacher z^* from the prior and then x from the channel. Splitting one factor of $p_F(x)$ off in Equation 6,

$$Z_n(F) = \mathbb{E}_{z^* \sim \text{ prior}, \, x \,\mid\, z^*, F}\left[\prod_{a=1}^{n}\int Dz^a \prod_i P_{\mathrm{out}}\big(x_i \mid (Fz^a)_i\big)\right]. \tag{42}$$

The replica index $a = 0$ plays the role of the **teacher** — it generates x via the leftover $p_F(x)$ factor. The remaining replicas $a = 1, ..., n$ are **students** drawn from the posterior $p(z \mid x, F) \propto p(z)\, P_{\mathrm{out}}(x|z, F)$ conditioned on the **same** observation.

Nishimori symmetry within a block. The joint density of $(z^0 := z^*, z^1, ..., z^n, x)$ given F factors as

$$p(z^0, ..., z^n, x \mid F) = \left(\prod_{a=0}^{n} p(z^a)\, P_{\mathrm{out}}(x \mid Fz^a)\right) / p_F(x)^n, \tag{43}$$

which is **manifestly symmetric** under arbitrary permutations of $(z^0, ..., z^n)$. This is the Nishimori identity in the matched teacher-student model: a posterior sample is statistically indistinguishable from the teacher itself, so the $n + 1$ replicas are exchangeable. Their pairwise overlaps therefore coincide,

$$\mathbb{E}[z^a \cdot z^b / D] = q^* \text{ for all } a \neq b, \tag{44}$$

and force $q_1 = q^*$ to equal the teacher-student magnetization. Within a block there is no separate "teacher-student" vs. "student-student" overlap; only one number survives.

Independence across blocks: $q_0 = 0$. It is tempting to argue simply that "teachers from different blocks are i.i.d. draws from a centered prior, so $\mathbb{E}\big[z^{*,\alpha} \cdot z^{*,\beta} / D\big] = 0$". That conclusion is correct but the premise is more subtle: the teachers' marginal under the replicated model is generically a **tilted** prior, and only collapses to the bare centered Gaussian at $n = 0$. Three independent ingredients are needed.

(i) Cross-block conditional independence given $\boldsymbol{F}$. The replicated weight factorises blockwise,

$$\rho\big(\{z^{(\alpha,a)}\}, \{x^\alpha\} \mid F\big) = \prod_\alpha \rho_\alpha\big(z^{(\alpha,0)}, ..., z^{(\alpha,n)}, x^\alpha \mid F\big), \quad \rho_\alpha \propto \prod_{a=0}^{n} p\big(z^{(\alpha,a)}\big)\, P_{\text{out}}\big(x^\alpha \mid F z^{(\alpha,a)}\big) \tag{45}$$

so cross-block expectations decouple into products of within-block marginals,

$$\mathbb{E}\big[z^{(\alpha,a)} \cdot z^{(\beta,b)}/D \mid F\big] = \mathbb{E}\big[z^{(\alpha,a)} \mid F\big]^\top \mathbb{E}\big[z^{(\beta,b)} \mid F\big]/D, \quad \alpha \neq \beta. \tag{46}$$

(ii) Within-block exchangeability. By the Nishimori symmetry just established, every replica in a block has the same marginal under ρ_α as the teacher: $\mathbb{E}\big[z^{(\alpha,a)} \mid F\big] = \mathbb{E}[z^{*,\alpha} \mid F]$ for every $a \in \{0, ..., n\}$.

(iii) Collapse of the teacher marginal to the bare prior at $\boldsymbol{n = 0}$. Integrating ρ_α over x^α and the other n replicas,

$$\rho_\alpha(z^{*,\alpha} \mid F) \propto p(z^{*,\alpha}) \int \mathrm{d}x^\alpha\, P_{\text{out}}(x^\alpha \mid F z^{*,\alpha})\, p_F(x^\alpha)^n. \tag{47}$$

At $n = 0$ the tilting factor $p_F(x)^n$ disappears, the x-integral is identically 1, and the marginal collapses to the bare centered prior $\mathcal{N}(0, I_D)$, **independently of $\boldsymbol{F}$**. Hence $\mathbb{E}[z^{*,\alpha} \mid F] = 0$ pointwise in F, and combined with (i)–(ii),

$$\mathbb{E}\big[z^{(\alpha,a)} \cdot z^{(\beta,b)}/D\big] = \mathbb{E}_F[0 \cdot 0/D] = 0, \quad \alpha \neq \beta, \tag{48}$$

which is precisely $q_0 = 0$.

The crucial role of $n = 0$ enters in step (iii): for $n \neq 0$ the marginal Equation 47 is a genuine F-dependent tilt of the prior, the within-block magnetization $\mathbb{E}[z^{*,\alpha} \mid F]$ does not collapse pointwise, and the cross-block average $q_0 = \mathbb{E}_F \|\mathbb{E}[z^{*,\alpha} \mid F]\|^2/D$ becomes the F-variance of the magnetization — generically nonzero.

Symmetric channels: $\boldsymbol{q_0 = 0}$ at any $\boldsymbol{n}$. The vanishing of q_0 extends to all n when the channel carries a $z \to -z$ symmetry: there exists an involution σ on the output alphabet such that

$$P_{\text{out}}(\sigma x \mid -y) = P_{\text{out}}(x \mid y) \quad \text{for all } y \in \mathbb{R}. \tag{49}$$

Examples: the sign channel of Section 4 satisfies this with $\sigma x = -x$, as do additive symmetric-noise channels (AWGN, symmetric Bernoulli flips, etc.). Combined with the centred prior $p(-z) = p(z)$, the substitution $z \to -z$ in the prior integral $p_F(x) = \int p(z)\, P_{\text{out}}(x \mid Fz)\, \mathrm{d}z$ yields $p_F(\sigma x) = p_F(x)$, and the same change of variables on Equation 47,

$$\rho_\alpha(-z^* \mid F) \propto p(-z^*) \int \mathrm{d}x\, P_{\text{out}}(x \mid -F z^*)\, p_F(x)^n \overset{x \to \sigma x}{=\!=\!=} p(z^*) \int \mathrm{d}x\, P_{\text{out}}(x \mid F z^*)\, p_F(x)^n \propto \rho_\alpha(z^* \mid F), \tag{50}$$

makes the within-block teacher marginal an **even** density in z^*, pointwise in F. Hence $\mathbb{E}[z^* \mid F] = 0$ identically, and steps (i)–(ii) give $q_0 = 0$ for every n. This is a symmetry, not a saddle accident — it explains why the noiseless sign channel of Section 4 exhibits $q_0 = 0$ also at $n = 1, 2, ...$ (cf. Section 4.3). It is, however, **not** a generic property: an asymmetric output channel breaks Equation 49 and reinstates $q_0 \neq 0$ at finite n.

When the collapse fails. The teacher-student rewriting is exact only at $n = 0$. For Rényi entropies at $n \neq 0$, Z_n probes the **tilted** measure $p_F(x)^{n+1}\, \mathrm{d}x$ rather than the marginal p_F: the extra factors $p_F(x)^n$ no longer admit an interpretation as posterior expectations against an underlying Bayes-optimal model, since $p_F(x)^n\, \mathrm{d}x$ is not a probability measure for $n \neq 0$

and there is no generative law in which x acts as an observation drawn from a teacher. In the present RS parametrization the Nishimori-type collapse at $n = 0$ reads

$$q_0 = 0, \quad q_1 = q, \quad \hat{q}_d = 0, \quad \hat{q}_0 = 0, \quad \hat{q}_1 = \hat{q}, \tag{51}$$

with $q, \hat{q}$ the surviving Bayes-optimal scalars (cf. Section APPENDIX D). These are properties of the order-n stationarity **at** $n = 0$, not of the system itself; in particular the standard Nishimori identity $\hat{m} = \hat{q}$ between teacher-student and student-student conjugate overlaps requires a parametrization that introduces those overlaps separately, whereas here teacher and student replicas have already been symmetrized inside the $(n+1)$-block. Consequently all five RS parameters $(q_0, q_1, \hat{q}_d, \hat{q}_0, \hat{q}_1)$ evolve independently in the general Rényi setting of Section 3, and the saddle has $q_0 \neq 0$ generically.

The saddle-point system reduces to

$$\boxed{\begin{cases} \hat{q} = \gamma\, \mathcal{J}(q) \\ q = \frac{\hat{q}}{1+\hat{q}} \end{cases}} \tag{52}$$

where

$$\mathcal{J}(q) := \mathbb{E}_{z\sim\mathcal{N}(0,q)}\left[\int \mathrm{d}x\, \frac{\left(\partial_z \tilde{\mathcal{P}}(x|z)\right)^2}{\tilde{\mathcal{P}}(x|z)}\right] \tag{53}$$

is the **Fisher information** of the channel $\tilde{\mathcal{P}}(\cdot\,|z)$ with respect to its conditioning variable z. Eliminating $\hat{q}$,

$$q = \frac{\gamma\, \mathcal{J}(q)}{1 + \gamma\, \mathcal{J}(q)}. \tag{54}$$

2.5.2 Final RS formula for the entropy

Putting together the output piece Equation 38 and the simplified prior piece, the quenched entropy density is

$$\boxed{s = \mathbb{E}_{z\sim\mathcal{N}(0,q^*)}\left[H\left(\tilde{\mathcal{P}}(\cdot \mid z)\right)\right] + \frac{1}{2\gamma}[-\log(1-q^*) - q^*],} \tag{55}$$

where q^* solves Equation 52 and $\tilde{\mathcal{P}}(x \mid z)$ is given by Equation 26 with $q_1 = q^*$. Equivalently, using $\log(1+\hat{q}^*) = -\log(1-q^*)$ on the saddle, the prior bracket reads $\log(1+\hat{q}^*) - q^*$. The expression is most cleanly obtained from the finite-n Rényi formula Equation 74: with the Bayes-optimal scaling $q_1, \hat{q}_1 = O(1)$ and $q_0, \hat{q}_0, \hat{q}_d = O(n)$, expansion of Equation 74 at order n gives

$$\mathcal{S}^{\mathrm{RS}}_{\mathrm{prior}}/m\,\Big|^{*}_{m\to 0} = n\left[-\frac{1}{2}\, q^*\hat{q}^* + \frac{1}{2}\,\hat{q}^* - \frac{1}{2}\log(1+\hat{q}^*)\right] + O(n^2), \tag{56}$$

and $s = -\gamma^{-1}\partial_{n[\ldots]} - \partial_n \Phi_n\,|_{n=0}$ together with $q^* = \frac{\hat{q}^*}{1+\hat{q}^*}$ produces the bracket above. The detailed expansion is collected in Section APPENDIX D.

The first term is the **conditional entropy** of x given the common Gaussian "signal" z of variance q^*; the residual variance $1 - q^*$ is integrated against the channel via w. The second term is the prior contribution: it is non-negative on $q^* \in [0, 1)$ (with $-\log(1-q^*) - q^* \geq 0$, vanishing only at $q^* = 0$) and diverges as $q^* \to 1$.

2.5.3 Consistency check: conditional entropy and mutual information

Decompose $H_F(X) = H_F(X \mid Z) + I_F(Z; X)$. The first term is **easy**: by independence across i and $(Fz)_i \to \mathcal{N}(0,1)$,

$$\lim_{N\to\infty} \frac{1}{N}\mathbb{E}_F H_F(X \mid Z) = \mathbb{E}_{h\sim\mathcal{N}(0,1)}[H(P_{\text{out}}(\cdot \mid h))]. \tag{57}$$

This equals the $q^* \to 1$ limit of the first term in Equation 55, where $\tilde{\mathcal{P}}(x|z) \to P_{\text{out}}(x|z)$. The replica calculation thus evaluates the mutual-information piece $\frac{1}{N}\mathbb{E}_F I_F(Z;X)$, which is captured by the prior bracket of Equation 55:

$$\frac{1}{N}\,\mathbb{E}_F I_F(Z;X) \to \frac{1}{2\gamma}\,[-\log(1-q^*) - q^*]. \tag{58}$$

This term is non-negative for $q^* \in [0,1)$ and **diverges** as $q^* \to 1$, consistent with the fact that for an informative channel the latent z becomes effectively decodable from x and the mutual information per site, while remaining $O(1/\gamma)$, picks up the unbounded $-\log(1-q^*)$ piece reflecting the perfect alignment of teacher and posterior. (The earlier statement that the mutual-information term vanishes at $q^* = 1$ was based on a spurious $(1-q^*)$ factor in the prior saddle and is incorrect.)

2.5.4 Replica symmetry: what is exact and what is an ansatz

The replicated action Equation 17 carries **two** distinct replica indices. The index $a = 0, \dots, n$ comes from the power $p_F(x)^{n+1}$ inside $Z_n(F)$. The index $\alpha = 1, \dots, m$ is introduced only to compute the quenched average

$$\mathbb{E}_F \log Z_n(F) = \lim_{m\to 0}\frac{1}{m}\log \mathbb{E}_F Z_n(F)^m. \tag{59}$$

Replica symmetry in these two directions has different meanings. The a-replicas form the internal “molecule” associated with one copy of Z_n, while the α-replicas are the usual quenched replicas of the partition function. A Parisi instability of the quenched free entropy should therefore first be looked for in the **outer** α-direction; internal breaking of the $(n+1)$-block is a secondary check.

Outer RS is essentially automatic at $n = 0$. The integer-replica object satisfies

$$Z_0(F) = \int \mathrm{d}x\, p_F(x) = 1 \quad \text{pointwise in } F. \tag{60}$$

Hence $Z_n(F) = 1 + nA_F + O(n^2)$ with $A_F = -H_F(X)$, and

$$\mathbb{E}_F Z_n(F)^m = 1 + mn\,\mathbb{E}_F A_F + O(n^2, m^2). \tag{61}$$

After $m \to 0$ and differentiation at $n = 0$, only the one-block term survives. Correlations among different disorder replicas can enter only at $O(n^2)$ — i.e. in higher cumulants of A_F — and therefore drop out of the Shannon derivative. Outer RS at $n = 0$ is thus not a physical assumption: it is forced/trivial at the order needed for the Shannon entropy.

$q_0 = 0$ from the one-block marginal. The vanishing of the RS inter-block overlap admits a direct derivation. For a single Z_n-block, the one-replica marginal is

$$\rho_n(z \mid F) = \frac{p(z)}{Z_n(F)}\int \mathrm{d}x\, P_{\text{out}}(x \mid Fz)\, p_F(x)^n, \tag{62}$$

so the RS cross-block overlap is

$$q_0(n) = \mathbb{E}_F \|\langle z\rangle_n(F)\|^2/D, \quad \langle z\rangle_n(F) := \int \mathrm{d}z\, z\, \rho_n(z \mid F). \tag{63}$$

At $n = 0$ the tilt $p_F(x)^n$ disappears, $\rho_0(z \mid F) = p(z)$, and the centred prior gives $|(z)_0(F) = 0$ pointwise in F, hence $q_0 = 0$. The conjugate variables in the present RS parametrization then satisfy

$$\hat{q}_0 = 0, \quad \hat{q}_d = 0, \quad \hat{q}_1 = \hat{q}, \tag{64}$$

rather than $\hat{q}_0 = \hat{q}_1$ (cf. Section APPENDIX D).

Inner RS as a Bayes-optimal ansatz. The remaining nontrivial assumption is the internal, Bayes-optimal RS ansatz inside one $(n+1)$-block. After singling out one factor of $p_F(x)$, the $a = 0$ replica plays the role of the teacher z^* and the others are posterior samples from $p(z \mid x, F)$. In the matched model the Nishimori identity makes teacher and posterior samples exchangeable, which equates the teacher–student and student–student overlap moments and justifies a single overlap q at the RS saddle. Exchangeability does **not**, however, by itself prove that the full overlap distribution is a delta — Parisi RSB ensembles are still exchangeable after averaging over the random hierarchy. The RS formula Equation 55 should therefore be understood as exact when the Nishimori saddle is stable against replicon/RSB perturbations. For the dense Gaussian HMM with simple log-concave or convex channels this is the expected situation; for more structured or hard channels the replicon (or a 1RSB perturbation) should be checked.

Finite Rényi order $n \neq 0$. The problem is no longer Bayes-optimal in the original sense: the measure is tilted by $p_F(x)^n$ and the Nishimori collapse is lost. The five-parameter RS saddle $(q_0, q_1, \hat{q}_d, \hat{q}_0, \hat{q}_1)$ is then a genuine ansatz, not a consequence of symmetry. If the channel has the parity symmetry Equation 49, the one-block marginal Equation 62 is even in z, hence $|(z)_n(F) = 0$ and the RS cross-block parameter satisfies

$$q_0(n) = 0 \quad \text{on the RS branch, for every } n. \tag{65}$$

This justifies the $q_0 = 0$ reduction for the sign channel (Section 4.3). It is, however, weaker than full outer RS: a parity-symmetric spin glass can still have a nontrivial overlap distribution. The correct claim is that **symmetry forces the RS q_0-branch to zero**, not that symmetry proves outer RS at all Rényi orders. Outer Parisi RSB at finite n remains a genuine question that requires a stability or interpolation argument; the appropriate first 1RSB ansatz, breaking the **outer** index α, is set up in Section APPENDIX E.

Near $\alpha = 1$. If the Shannon RS saddle is locally stable, RS persists for α in a neighbourhood of 1 by continuity. Far from $\alpha = 1$ — especially as $\alpha \to 0$ or $\alpha \to \infty$, where the Rényi entropy probes rare support/peak structure — outer RSB and large-deviation effects become more plausible and a stability check is needed.

The clean message, summarized:

$$\boxed{\text{“Shannon: outer RS is exact/irrelevant; inner RS is Nishimori-stable if the Bayes saddle is stable.”}} \tag{66}$$

$$\boxed{\text{“Finite Rényi: RS is an ansatz; parity can force “ q\_0 = 0 “ but does not prove full RS.”}} \tag{67}$$

3 Quenched Rényi entropy at finite n

3.1 Setup

The same replica machinery yields the **quenched Rényi entropy** of order $\alpha = n + 1$ (with $\alpha > 0$, $\alpha \neq 1$),

$$S_\alpha = \mathbb{E}_F[H_\alpha(X)], \quad H_\alpha(X) = \frac{1}{1-\alpha} \log \int \mathrm{d}x\, p_F(x)^\alpha = -\frac{1}{n} \log Z_n(F), \tag{68}$$

with $Z_n(F)$ exactly the integer-replica object defined in Equation 6. The Shannon entropy of Section 2 is recovered as $\alpha \to 1$ ($n \to 0$) since $-n^{-1} \log Z_n \to -\partial_n \log Z_n\,|_{n=0}$.

The quenched average still requires the second replica trick $\mathbb{E}_F \log Z_n = \lim_{m\to 0} m^{-1} \log \mathbb{E}_F[Z_n^m]$, so steps 2.2–2.3 carry over verbatim: the disorder-averaged action Equation 17, the Hubbard–Stratonovich step, and the $m \to 0$ expansion of Equation 32 are unchanged. The only difference is that we keep n finite instead of differentiating at $n = 0$. Defining

$$\varphi_n(q_0, q_1, \hat{q}_d, \hat{q}_0, \hat{q}_1) := \mathcal{S}^{\mathrm{RS}}_{\mathrm{prior}}/m\,\big|_{m\to 0} + \gamma\, \Phi_n(q_0, q_1) \tag{69}$$

exactly as in Equation 29 and Equation 32, the Rényi entropy density is

$$\boxed{s_\alpha = \lim_{N\to\infty} \frac{S_\alpha}{N} = -\frac{1}{n\gamma} \operatorname*{extr}_{q_0,q_1,\hat{q}_d,\hat{q}_0,\hat{q}_1} \varphi_n.} \tag{70}$$

3.2 RS saddle-point equations

Stationarity of φ_n in the conjugate variables $(\hat{q}_d, \hat{q}_0, \hat{q}_1)$ involves only the prior piece. Differentiating Equation 32 and combining the resulting three relations yields the matrix-inversion identities $Q = \left(I - \hat{Q}\right)^{-1}$ on each RS subspace (cf. Section APPENDIX A):

$$\boxed{\begin{cases} 1 - q_1 = \frac{1}{A} \\ 1 + nq_1 - (n+1)q_0 = \frac{1}{B_0} \\ q_0 = \frac{\hat{q}_0}{B_0^2} \end{cases}} \tag{71}$$

with

$$A := 1 - \hat{q}_d + \hat{q}_1, \quad B_0 := 1 - \hat{q}_d - n\hat{q}_1 + (n+1)\hat{q}_0. \tag{72}$$

Stationarity in (q_0, q_1) couples the prior to the output term and gives the conjugate parameters in terms of the channel (the prior pieces in Equation 32 contribute $-(n+1)^2 q_0 \hat{q}_0/2$ and $-n(n+1)q_1\hat{q}_1/2$, which are linear in q_0, q_1):

$$\boxed{\begin{cases} \hat{q}_1 = \frac{2\gamma}{n(n+1)}\, \partial_{q_1} \Phi_n(q_0, q_1) \\ \hat{q}_0 = -\frac{2\gamma}{(n+1)^2}\, \partial_{q_0} \Phi_n(q_0, q_1). \end{cases}} \tag{73}$$

Equations Equation 71 and Equation 73 form a closed five-dimensional system. In the Bayes-optimal Shannon limit $n \to 0$, Nishimori symmetry collapses $(q_0, \hat{q}_0) \to (0, 0)$, the second relation in Equation 71 reduces to $1 + nq_1 = \frac{1}{B_0}$, and the system contracts to Equation 52 after the linear-in-n expansion of Section APPENDIX D. At finite n the Nishimori symmetry is broken and all five parameters are generically nonzero.

3.3 Final RS formula

Let $(q_0^*, q_1^*, \hat{q}_d^*, \hat{q}_0^*, \hat{q}_1^*)$ denote a solution of Equation 71 and Equation 73. Substituting Equation 71 into Equation 32 — using $A^* = 1/(1 - q_1^*)$, $B_0^* = 1/(1 + nq_1^* - (n+1)q_0^*)$, and $\hat{q}_d^* = -q_1^*/(1 - q_1^*) + \hat{q}_1^*$ — the prior contribution at the saddle reads

$$
\begin{aligned}
\mathcal{S}^{\mathrm{RS}}_{\mathrm{prior}}/m \,\big|^{*}_{m\to 0} = &\frac{n+1}{2}\,\frac{q_1^*}{1-q_1^*} + \frac{n}{2}\log(1-q_1^*) \\
&+\frac{1}{2}\log(1+nq_1^* - (n+1)q_0^*) + \frac{n+1}{2}\,\frac{q_0^*}{1+nq_1^*-(n+1)q_0^*} \\
&-(n+1)\frac{1+nq_1^*}{2}\,\hat{q}_1^* + \frac{(n+1)^2}{2}\,q_0^*\hat{q}_0^*.
\end{aligned}
\tag{74}
$$

Together with $\gamma\,\Phi_n^*$ from the output piece, the Rényi entropy density is

$$
\boxed{s_\alpha = -\frac{1}{n\gamma}\Big[\mathcal{S}^{\mathrm{RS}}_{\mathrm{prior}}/m \,\big|^{*}_{m\to 0} + \gamma\,\Phi_n^*\Big], \quad \alpha = n+1,} \tag{75}
$$

with Equation 74 supplying the prior part. The output integral

$$
\Phi_n^* = \int Du \log \int Dv \int \mathrm{d}x \,\Big[\tilde{\mathcal{P}}\Big(x \;\Big|\; \sqrt{q_0^*}u + \sqrt{q_1^*-q_0^*}v\Big)\Big]^{n+1} \tag{76}
$$

is the natural generalization of the conditional output entropy of Equation 38: as $n \to 0$ it tends to zero linearly with slope equal to minus the conditional entropy at common-cause variance q_1^*, recovering the first term of Equation 55.

For **integer** $n \geq 1$ the output integral Φ_n reduces to a finite-dimensional Gaussian integral with $(n+1)$ explicit replica fields w^a, and Equation 75 gives the **integer-order Rényi entropies** $H_2, H_3, \ldots$. The collision entropy H_2 $(n=1)$ is particularly tractable since

$$
\Phi_1(q_0, q_1) = \int Du \log \int Dv \int \mathrm{d}x\, \tilde{\mathcal{P}}(x \mid z(u,v))^2, \tag{77}
$$

which is purely a Gaussian double integral against the squared channel.

3.4 Boundary cases

The two extreme limits $\alpha \to \infty$ (min-entropy) and $\alpha \to 0$ (Hartley entropy) probe complementary aspects of p_F — the height of the peak and the size of the support — and are most cleanly analyzed by going back to $Z_n(F)$ rather than through Equation 75.

3.4.1 Min-entropy limit ($\alpha \to \infty$, $n \to \infty$)

The Rényi entropy concentrates on the most probable output:

$$
H_\infty(X) := -\log \sup_x p_F(x) = \lim_{\alpha\to\infty} H_\alpha(X). \tag{78}
$$

At the level of Z_n, Laplace asymptotics give

$$
Z_n(F) \sim \Big[\sup_x p_F(x)\Big]^{n+1} e^{O(\log n)} \quad (n \to \infty), \tag{79}
$$

so $-n^{-1}\log Z_n \to H_\infty$ as expected. Inside the replica formula, the **pointwise** Laplace step at fixed latent z is correct,

$$
\int \mathrm{d}x\, \Big[\tilde{\mathcal{P}}(x|z)\Big]^{n+1} = \Big[\sup_x \tilde{\mathcal{P}}(x|z)\Big]^{n+1} e^{O(\log n)}, \tag{80}
$$

but the inner v-integral inside $\Phi_n = \int Du \log \int Dv\,(\ldots)$ is itself dominated by Laplace at $n \to \infty$. Writing $g(u,v) := \log\sup_x \tilde{\mathcal{P}}(x \mid z(u,v))$, the leading large-$n$ behaviour is

$$
\frac{1}{n+1}\,\Phi_n(q_0^*, q_1^*) \to \int Du \sup_v g(u,v) - \frac{1}{2(n+1)}\,v_*^2 + \ldots, \tag{81}
$$

and the supremum over v at fixed u generically does **not** reduce to a Gaussian average over $z \sim \mathcal{N}(0, q_1^\infty)$. A correct $\alpha \to \infty$ treatment therefore requires a joint Laplace/large-deviation analysis over both x and v (and stabilization of the conjugate parameters $\hat{q}^*_{d,0,1}$). Skipping the v-Laplace step would give the heuristic expression

$$s_\infty^{\text{heur.}} = \mathbb{E}_{z\sim\mathcal{N}(0,q_1^\infty)}\Big[H_\infty\big(\tilde{\mathcal{P}}(\cdot \mid z)\big)\Big], \tag{82}$$

the expected conditional min-entropy of $\tilde{\mathcal{P}}(\cdot\,|z)$ at the limiting overlap. This is the natural $\alpha = \infty$ analogue of Equation 55, but it should be regarded as a **heuristic estimate** rather than a derivation: it is exact only in regimes where $\sup_v g(u,v)$ collapses onto a Gaussian average over v (e.g. when g is a quadratic function of $z(u,v)$, or when the channel-induced fluctuations dominate over the Gaussian prior weight). The Rényi monotonicity $s_\infty \le s$ is in any case preserved. When the channel is sufficiently informative one expects $q_1^\infty \to 1$ (so $\tilde{\mathcal{P}} \to P_{\text{out}}$ and $z \sim \mathcal{N}(0,1)$), recovering the natural form $s_\infty = \mathbb{E}_{z\sim\mathcal{N}(0,1)}[H_\infty(P_{\text{out}}(\cdot\,|z))]$.

3.4.2 Hartley entropy ($\alpha \to 0$, $n \to -1$)

The opposite extreme is the **Hartley entropy**

$$H_0(X) := \log|\text{supp}(p_F)| = \lim_{\alpha\to0^+} H_\alpha(X), \tag{83}$$

which measures the size of the achievable output set rather than its peakedness. Setting $n = -1$ in Equation 6 gives $Z_{-1}(F) = |\text{supp}(p_F)|$ directly. For continuous channels with $p_F > 0$ on $\mathbb{R}$ (e.g. Gaussian P_{out}), $H_0 = +\infty$ trivially; the meaningful regime is channels with restricted support — most naturally, **discrete-output** channels.

Canonical example: hard-threshold (sign) channel

Take $P_{\text{out}}(x \mid h) = \delta_{x,\,\text{sign}(h)}$ with $x \in \{-1,+1\}$, so that

$$p_F(x) = \int Dz \prod_{i=1}^{N} \Theta(x_i(Fz)_i), \tag{84}$$

and

$$Z_{-1}(F) = \big|\{y \in \{\pm1\}^N : \exists z \text{ with } \text{sign}(Fz) = y\}\big| =: T(F) \tag{85}$$

counts the **sign patterns** realizable by the N random hyperplanes $\{z : (Fz)_i = 0\}$ in $\mathbb{R}^D$.

By Cover's hyperplane-counting theorem (Cover, 1965; equivalently, Wendel's formula, 1962), for F with rows in **general position** — which holds **almost surely** for F with a continuous (in particular Gaussian) distribution — the count $T(F)$ is **deterministic** and equals

$$T(F) = T(N,D) := 2\sum_{k=0}^{D-1}\binom{N-1}{k} \quad \text{a.s.} \tag{86}$$

No expectation or concentration step is needed: the statement that $T(F)$ depends on F only through general position is a geometric fact, not a typicality fact. Using $\sum_{k=0}^{D-1}\binom{N-1}{k} \sim \binom{N-1}{D-1} \sim e^{Nh_2(1/\gamma)}$ for $D-1 < \frac{N-1}{2}$, one obtains the celebrated **Cover dichotomy**:

$$\boxed{s_0 = \lim_{N\to\infty}\frac{\log T(N,D)}{N} = \begin{cases}\log 2 & \gamma \le 2 \quad \text{(linearly-separable phase)}\\ h_2(1/\gamma) & \gamma > 2 \quad \text{(over-determined phase)}\end{cases}} \tag{87}$$

where

$$h_2(p) := -p\log p - (1-p)\log(1-p). \tag{88}$$

Below the Cover threshold $\gamma_c = 2$ all 2^N sign patterns are typically realized (so the support is generic and $s_0 = \log 2$ saturates the trivial upper bound); above it the achievable fraction shrinks as $h_2(1/\gamma)/\log 2$, and $s_0 \to 0$ as $\gamma \to \infty$. The same result is recovered from the $n \to -1$ saddle-point analysis of Equation 6 — the **Gardner-volume** calculation in its earliest, replica-symmetric form — in agreement with the general Rényi formula Equation 70.

3.4.3 Summary of limits

Combining the boundary cases with the formulas of Section 2 and Section 3:

- $\alpha = 1$ (Shannon): $s = \mathbb{E}_z\left[H\left(\tilde{\mathcal{P}}(\cdot\,|z)\right)\right]$ + prior overlap term, see Equation 55.
- $\alpha = 2$ (collision): $s_2 = -\gamma^{-1}\frac{\varphi_1^*}{1} = -\varphi_1^*/\gamma$, with $\Phi_1(q_0, q_1) = \int Du \log \int Dv \int \mathrm{d}x\, \tilde{\mathcal{P}}^2$.
- $\alpha \to \infty$ (min): $s_\infty = \mathbb{E}_z\left[H_\infty\left(\tilde{\mathcal{P}}(\cdot\,|z)\right)\right]$, see Equation 82.
- $\alpha \to 0$ (Hartley): $s_0 = \log|\mathrm{supp}|$, channel-dependent (sign channel: Equation 87).

The chain $s_0 \geq s \geq s_2 \geq s_\infty$ encodes the ordering of moments of p_F.

4 Worked example: noiseless sign channel

4.1 Setup

Take the deterministic sign channel

$$P_{\text{out}}(x \mid h) = \delta_{x,\,\text{sign}(h)}, \quad x \in \{-1, +1\}. \tag{89}$$

This is the same channel that produced the Cover dichotomy at $\alpha = 0$ in Equation 87; here we work it out across the full Rényi profile. Throughout this section let

$$\Phi(t) := \int_{-\infty}^{t} \frac{e^{-s^2/2}}{\sqrt{2\pi}}\,\mathrm{d}s, \quad \varphi(t) := \Phi'(t) = \frac{e^{-t^2/2}}{\sqrt{2\pi}}, \tag{90}$$

denote the standard normal CDF and density (not to be confused with the action φ_n and the integral Φ_n of Section 2 — both carry a subscript).

The effective channel Equation 26 collapses to the **probit**

$$\tilde{\mathcal{P}}(x \mid z) = \Pr_{w \sim \mathcal{N}(0,1)}\left[\mathrm{sign}\left(z + \sqrt{1-q_1}\,w\right) = x\right] = \Phi\left(xz/\sqrt{1-q_1}\right). \tag{91}$$

A hard threshold convolved with a Gaussian of variance $1 - q_1$: as $q_1 \to 1$ the threshold sharpens onto the deterministic channel; as $q_1 \to 0$ the noise washes it out and $\tilde{\mathcal{P}} \to 1/2$.

4.2 Shannon entropy $(\alpha = 1)$

4.2.1 Conditional output entropy

The output entropy at fixed z is the binary entropy of Equation 91:

$$H\left(\tilde{\mathcal{P}}(\cdot\,|z)\right) = h_2\left(\Phi\left(z/\sqrt{1-q_1}\right)\right). \tag{92}$$

Substituting in Equation 38 and changing variable to $t = z/\sqrt{1-q_1}$ (which is $\mathcal{N}(0, q_1/(1-q_1))$ when $z \sim \mathcal{N}(0, q_1)$),

$$\partial_n \Phi_n\,|_{n=0} = -\mathbb{E}_{t \sim \mathcal{N}(0,\,q_1/(1-q_1))}[h_2(\Phi(t))]. \tag{93}$$

This is monotonically decreasing in q_1: at $q_1 = 0$ the latent t is a point mass at 0 and $h_2(\Phi(0)) = \log 2$; at $q_1 = 1$ the variance of t diverges, $|t|$ is typically large, $\Phi(t) \in \{0, 1\}$ a.s., and the conditional entropy collapses to 0.

4.2.2 Fisher information

From Equation 91, $\partial_z \tilde{\mathcal{P}}(x|z) = (x/\sqrt{1-q})\,\varphi(z/\sqrt{1-q})$. Summing $\left(\partial_z \tilde{\mathcal{P}}\right)^2/\tilde{\mathcal{P}}$ over $x \in \{-1, +1\}$ and using $1/\Phi + 1/(1-\Phi) = 1/(\Phi(1-\Phi))$,

$$\mathcal{I}(q) = \frac{1}{1-q}\,\mathbb{E}_{z \sim \mathcal{N}(0,q)}\left[\frac{\varphi^2(z/\sqrt{1-q})}{\Phi(z/\sqrt{1-q})\,(1 - \Phi(z/\sqrt{1-q}))}\right]. \tag{94}$$

The change of variable $t = z/\sqrt{1-q}$ rewrites this as a one-dimensional Gaussian quadrature against $\mathcal{N}(0, q/(1-q))$:

$$\mathcal{J}(q) = \frac{1}{1-q}\,\mathbb{E}_{t\sim\mathcal{N}(0,\,q/(1-q))}\left[\frac{\varphi^2(t)}{\Phi(t)(1-\Phi(t))}\right]. \tag{95}$$

Two reference values:

- $\boldsymbol{q = 0}$**:** $t = 0$ a.s., so $\varphi^2(0)/(\Phi(0)(1-\Phi(0))) = (1/(2\pi))/(1/4) = 2/\pi$, hence $\mathcal{J}(0) = 2/\pi$.
- $\boldsymbol{q \to 1}$**:** using the Mills tail $\Phi(t)(1-\Phi(t)) \sim \varphi(t)/|t|$ for $|t| \to \infty$, the integrand grows like $|t|\,\varphi(t)$ in the tails. The bounded integral $C := \int_{\mathbb{R}} \varphi^2(t)/(\Phi(t)(1-\Phi(t)))\,\left(\mathrm{d}t/\sqrt{2\pi}\right)$ converges, and a saddle estimate of Equation 95 gives $\mathcal{J}(q) \sim C/\sqrt{q(1-q)} \sim C/\sqrt{1-q}$.

4.2.3 Saddle point and final formula

The Bayes-optimal saddle Equation 52 reads

$$q = \frac{\gamma\,\mathcal{J}(q)}{1+\gamma\,\mathcal{J}(q)}, \quad \hat{q} = \gamma\,\mathcal{J}(q), \tag{96}$$

with $\mathcal{J}(q)$ given by Equation 95. The two extreme regimes are:

- **Small** $\boldsymbol{\gamma}$**:** $\mathcal{J}(0) = 2/\pi$, so $\hat{q}^* \sim 2\gamma/\pi$ and $q^* \sim 2\gamma/\pi \to 0$. The latent code is essentially invisible to the output and the entropy density tends to $\log 2$.
- **Large** $\boldsymbol{\gamma}$**:** combining the saddle with $\mathcal{J}(q) \sim C/\sqrt{1-q}$ gives $\hat{q}^* \sim \gamma C/\sqrt{1-q^*}$, and $1-q^* = 1/(1+\hat{q}^*) \sim 1/\hat{q}^*$, hence $\sqrt{1-q^*} \sim (\gamma C)^{-1}$, i.e.
$$1 - q^* \sim (\gamma C)^{-2}. \tag{97}$$
So $q^* \to 1$ slowly. The conditional-entropy term in Equation 55 vanishes (as $h_2(\Phi(t))$ averaged against a Gaussian of diverging variance) and the prior bracket reduces to $-\log(1-q^*) - q^* \sim 2\log(\gamma C) - 1$, so $s \to 0$ at rate $\log(\gamma)/\gamma$.

The replica-symmetric Shannon entropy density is, from Equation 55,

$$\boxed{s = \mathbb{E}_{t\sim\mathcal{N}(0,\,q^*/(1-q^*))}[h_2(\Phi(t))] + \frac{1}{2\gamma}[-\log(1-q^*) - q^*],} \tag{98}$$

with q^* solving Equation 96.

The outer-1RSB analysis of the sign channel — Shannon-order reduction to the RS saddle Equation 98, with the symmetry-induced collapse $q_0 = 0$ and a remarks on outer-replicon stability at finite Rényi order — is worked out in Section E.7. We do **not** prove full absence of outer 1RSB at all α: parity symmetry forces $q_0 = 0$ on the RS branch but leaves the cross-group overlap q_{s} to be checked by a replicon computation, which we do not carry out here. The only non-trivial RSB-like signature in the Rényi profile of this channel that we positively identify sits at $\alpha = 0$, namely the Cover dichotomy Equation 87, which counts achievable output patterns rather than measuring posterior structure.

4.3 Generic Rényi order ($q_0 = 0$ reduction)

Numerical solution of the five-dimensional system Equation 71–Equation 73 shows that $q_0^* = 0$ for the sign channel across all tested values of n and γ. We exploit this observation — verified a posteriori at the end of this section — to reduce the saddle to a single scalar equation.

4.3.1 Simplified output integral

With $q_0 = 0$ the signal field $\sqrt{q_0}u$ in Equation 25 drops out: the output integral Equation 29 loses its u-dependence and the outer Du integral is trivial,

$$\Phi_n(0, q_1) = \log \int Dv \int \mathrm{d}x \left[\tilde{\mathcal{P}}(x \mid \sqrt{q_1}\, v)\right]^{n+1}. \tag{99}$$

For the sign channel, inserting Equation 91 and changing variable to $t = v\sqrt{q_1/(1-q_1)} \sim \mathcal{N}(0, \rho)$ with $\rho := q_1/(1-q_1)$,

$$\boxed{\Phi_n(0, q_1) = \log \mathbb{E}_{t\sim\mathcal{N}(0,\rho)}[\Phi(t)^{n+1} + (1-\Phi(t))^{n+1}], \quad \rho := \frac{q_1}{1-q_1}.} \tag{100}$$

4.3.2 Reduced saddle-point equations

Setting $q_0 = 0$ in the third line of Equation 71 immediately gives $\hat{q}_0 = q_0\, B_0^2 = 0$, confirming self-consistency. The first two lines then read

$$A = \frac{1}{1-q_1}, \quad B_0 = \frac{1}{1+nq_1}. \tag{101}$$

Using $A = 1 - \hat{q}_d + \hat{q}_1$ and $B_0 = 1 - \hat{q}_d - n\hat{q}_1$, subtracting gives

$$(n+1)\hat{q}_1 = \frac{1}{1-q_1} - \frac{1}{1+nq_1} = \frac{(n+1)q_1}{(1-q_1)(1+nq_1)}, \tag{102}$$

hence

$$\hat{q}_1^* = \frac{q_1^*}{(1-q_1^*)(1+nq_1^*)}. \tag{103}$$

Substituting into the first line of Equation 73 (with $\hat{q}_0 = 0$) yields a **single** self-consistency equation for q_1:

$$\boxed{\frac{q_1}{(1-q_1)(1+nq_1)} = \frac{2\gamma}{n(n+1)}\, \partial_{q_1} \Phi_n(0, q_1).} \tag{104}$$

The five-dimensional system Equation 71–Equation 73 thus reduces to a single one-dimensional root-finding problem.

4.3.3 Simplified prior and final formula

Inserting $q_0^* = 0$, $\hat{q}_0^* = 0$, and Equation 103 into Equation 74, the $(n+1)q_1^*/(2(1-q_1^*))$ terms cancel exactly and the prior collapses to

$$\boxed{\mathcal{S}^{\mathrm{RS}}_{\mathrm{prior}}/m \,\Big|^{*}_{m\to 0} = \frac{n}{2}\log(1-q_1^*) + \frac{1}{2}\log(1+nq_1^*).} \tag{105}$$

Together with Equation 100 and the general Rényi formula Equation 75, the entropy density is

$$\boxed{s_\alpha = -\frac{\log(1-q_1^*)}{2\gamma} - \frac{\log(1+nq_1^*)}{2n\gamma} - \frac{1}{n}\log \mathbb{E}_{t\sim\mathcal{N}(0,\rho^*)}[\Phi(t)^{n+1} + (1-\Phi(t))^{n+1}],} \tag{106}$$

with $\rho^* = q_1^*/(1-q_1^*)$ and q_1^* the positive root of Equation 104.

Shannon limit. As $n \to 0$ $(\alpha \to 1)$: $\log(1+nq_1^*)/(2n\gamma) \to q^*/(2\gamma)$, and expanding the log-expectation at leading order in n gives $-n\,\mathbb{E}_t[h_2(\Phi(t))]$, so that $-1/n$ times the log recovers $\mathbb{E}_t[h_2(\Phi(t))]$. The formula Equation 106 then reduces to Equation 98. The saddle Equation 104 similarly contracts to Equation 96 via $\frac{\log(1+nq_1)}{n} \to q_1$ and the heat-equation identity $\partial_{q_1}\Phi_{n(0,q_1)}|_{n\to 0} = -(n/2)\,\mathcal{I}(q_1) + O(n^2)$.

4.4 Collision entropy $(\alpha = 2)$

For binary outputs,

$$\sum_x \tilde{\mathcal{P}}(x \mid z)^2 = \Phi(\xi)^2 + (1-\Phi(\xi))^2 = 1 - 2\Phi(\xi)(1-\Phi(\xi)), \quad \xi := z/\sqrt{1-q_1}. \tag{107}$$

Inserting in Φ_1 from Equation 29 at $n = 1$,

$$\Phi_1(q_0, q_1) = \int Du \log \int Dv \, [1 - 2\Phi(\xi(u,v))(1 - \Phi(\xi(u,v)))], \tag{108}$$

$$\xi(u,v) := (\sqrt{q_0}\, u + \sqrt{q_1 - q_0}\, v)/\sqrt{1-q_1}, \tag{109}$$

a two-dimensional Gaussian quadrature, easily evaluated numerically. The collision-entropy saddle then comes from Equation 71 and Equation 73 at $n = 1$, and

$$s_2 = -\frac{1}{\gamma} \varphi_1 \,|^* \tag{110}$$

from Equation 75. Unlike the Shannon case, Nishimori symmetry is broken at $n = 1$, so all five RS parameters $(q_0^*, q_1^*, \hat{q}_d^*, \hat{q}_0^*, \hat{q}_1^*)$ are generically nonzero. Under the $q_0 = 0$ assumption of Section 4.3, which is observed to hold numerically also at $n = 1$, the formula Equation 106 specialises to

$$s_2 = -\frac{\log(1-q_1^*)}{2\gamma} - \frac{\log(1+q_1^*)}{2\gamma} - \log \mathbb{E}_{t \sim \mathcal{N}(0,\rho^*)}[1 - 2\Phi(t)(1-\Phi(t))], \tag{111}$$

with q_1^* solving Equation 104 at $n = 1$, which simplifies to a single integral equation.

4.5 Min-entropy $(\alpha \to \infty)$

For binary outputs and $z \neq 0$,

$$\sup_x \tilde{\mathcal{P}}(x \mid z) = \max\Big(\Phi\big(z/\sqrt{1-q_1}\big), \Phi\big(-z/\sqrt{1-q_1}\big)\Big) = \Phi\big(|z|/\sqrt{1-q_1}\big), \tag{112}$$

so $H_\infty\big(\tilde{\mathcal{P}}(\cdot \mid z)\big) = -\log \Phi\big(|z|/\sqrt{1-q_1}\big)$. Inserting into the **heuristic** min-entropy formula Equation 82 (subject to the caveat of Section 3.4.1 — the v-Laplace step is not justified in general), the channel's $z \to -z$ symmetry gives

$$s_\infty^{\text{heur.}} = -2 \int_0^\infty \frac{e^{-z^2/(2q_1^\infty)}}{\sqrt{2\pi q_1^\infty}} \log \Phi\big(z/\sqrt{1-q_1^\infty}\big)\, \mathrm{d}z. \tag{113}$$

At $q_1^\infty \to 0$ the integrand reduces to $-\log \Phi(0) = \log 2$ (uninformative limit); at $q_1^\infty \to 1$ the integrand sharpens onto $z > 0$ where $\Phi \to 1$, hence $s_\infty^{\text{heur.}} \to 0$. A rigorous min-entropy analysis for the sign channel would require a separate large-deviation treatment of the joint (v, x) supremum at fixed u inside Equation 29.

4.6 Hartley entropy $(\alpha \to 0)$: Cover dichotomy

The $\alpha \to 0$ result was already derived for this channel in Section 3.4; for completeness we recall

$$s_0 = \begin{cases} \log 2 & \gamma \le 2 \quad \text{(linearly separable phase)} \\ h_2(1/\gamma) & \gamma > 2 \quad \text{(over-determined phase)} \end{cases} \tag{114}$$

the **Cover dichotomy** of Equation 87. Below the threshold $\gamma_c = 2$ all 2^N sign patterns are typically realized and $s_0 = \log 2$ saturates; above it, the achievable fraction shrinks with the binary entropy of $1/\gamma$.

4.7 Summary

The noiseless sign channel admits closed-form expressions (modulo at most a two-dimensional Gaussian quadrature) at every Rényi order considered, with the **only** non-trivial scalar input being the Bayes-optimal overlap $q^* = q^*(\gamma)$ from Equation 96 entering the Shannon case.

The Rényi profile satisfies $s_0 \geq s \geq s_2 \geq s_\infty$, with all four collapsing onto $\log 2$ as $\gamma \to 0$ (uninformative regime). For $\gamma \to \infty$ the orderings $s, s_2, s_\infty \to 0$, while s_0 retains the explicit Cover form $h_2(1/\gamma)$ that decays only logarithmically — a faithful diagnostic that, even when typical outputs concentrate, the **support** of p_F remains exponentially rich. Figure 1 shows the full RS Rényi entropy profile across $\alpha \in [0.02, 10]$ obtained by numerically solving the saddle-point system, sandwiched between the Cover envelope at $\alpha \to 0$ and the min-entropy limit at $\alpha \to \infty$.

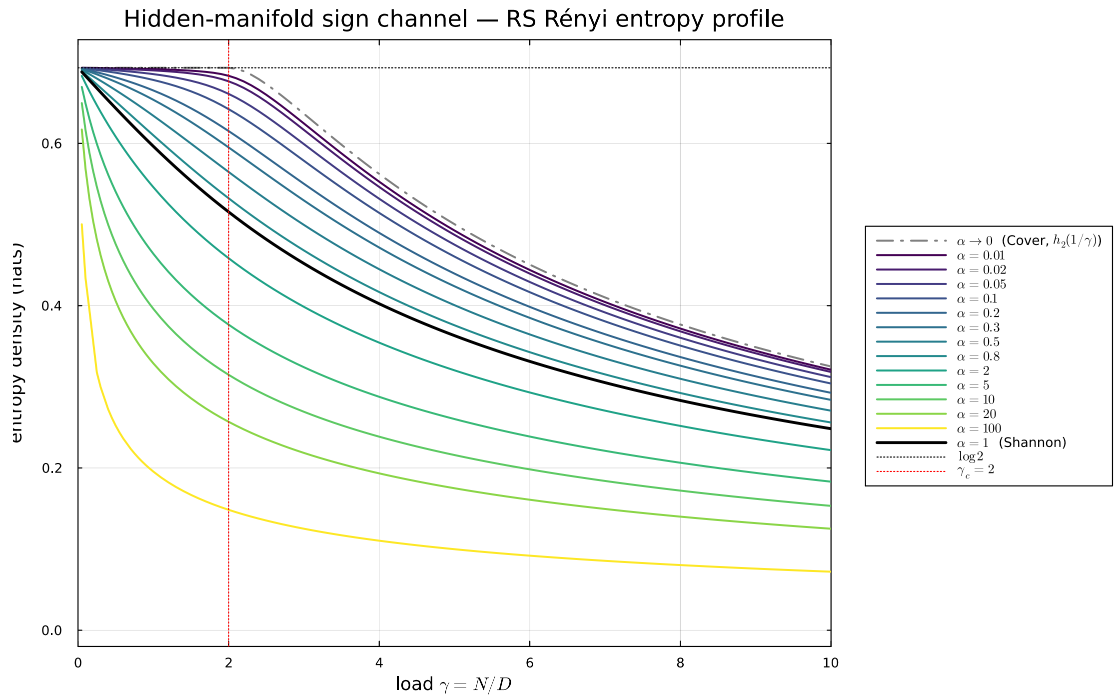

Figure 1: Replica-symmetric Rényi entropy density $s_{\alpha(\gamma)}$ for the hidden-manifold sign channel as a function of the load $\gamma = N/D$, for $\alpha \in \{0.02, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.9, 1, 2, 5, 10\}$ (color-coded by $\log \alpha$, viridis from purple to yellow). The black curve is the Shannon entropy ($\alpha = 1$, Equation 98) and the gray dash-dot envelope is the Cover/Hartley limit $s_0 = h_2(1/\gamma)$ for $\gamma > 2$, $s_0 = \log 2$ otherwise (Equation 87). Curves with $\alpha \neq 1$ are produced by the $q_0 = 0$ reduction of Section 4.3 (for $\alpha \in \{0.4, 0.5, 0.9\}$ the full five-parameter saddle is used; the two ansätze agree to numerical precision wherever both have been computed). Rényi monotonicity $\partial_\alpha s_\alpha \leq 0$ is satisfied across the entire grid; all curves collapse onto $\log 2$ as $\gamma \to 0$.

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APPENDIX A Spectrum of $\hat{Q}$ under the RS ansatz

Decompose $\mathbb{R}^M$ into three invariant subspaces of $\hat{Q}$:

1. **Within-block antisymmetric:** $\sum_a v^{\alpha a} = 0$ for every α. Dimension mn. On such v,
$$\left(\hat{Q}v\right)^{\alpha a} = \hat{q}_d\, v^{\alpha a} + \hat{q}_1 \sum_{b \neq a} v^{\alpha b} + \hat{q}_0 \sum_{\beta \neq \alpha} \sum_b v^{\beta b} = (\hat{q}_d - \hat{q}_1)\, v^{\alpha a}. \tag{115}$$
Eigenvalue $\lambda_1 = \hat{q}_d - \hat{q}_1$, multiplicity mn.
2. **Between-block antisymmetric:** $v^{\alpha a} = c_\alpha$ with $\sum_\alpha c_\alpha = 0$. Dimension $m-1$.
$$\left(\hat{Q}v\right)^{\alpha a} = (\hat{q}_d + n\hat{q}_1)c_\alpha + (n+1)\hat{q}_0 \sum_{\beta \neq \alpha} c_\beta = [\hat{q}_d + n\hat{q}_1 - (n+1)\hat{q}_0]c_\alpha. \tag{116}$$
Eigenvalue $\lambda_2 = \hat{q}_d + n\hat{q}_1 - (n+1)\hat{q}_0$, multiplicity $m-1$.
3. **Fully symmetric:** $v^{\alpha a} \equiv \text{const}$. Dimension 1. Eigenvalue
$$\lambda_3 = \hat{q}_d + n\hat{q}_1 + (m-1)(n+1)\hat{q}_0. \tag{117}$$

Dimensions add to $mn + (m-1) + 1 = m(n+1) = M$, and Equation 22 follows. Direct enumeration of M^2 pairs gives Equation 24.

APPENDIX B The $m \to 0$ expansion of $\log\det\left(I - \hat{Q}\right)$

With A, B, C as in the main text, $\log\det\left(I - \hat{Q}\right) = mn \log A + (m-1)\log B + \log C$. Note $C - B = -m(n+1)\hat{q}_0$, so for small m,
$$\log C = \log B + \log(1 - m(n+1)\hat{q}_0/B) = \log B - \frac{m(n+1)\hat{q}_0}{B} + O(m^2). \tag{118}$$
Hence
$$(m-1)\log B + \log C = m \log B - \frac{m(n+1)\hat{q}_0}{B} + O(m^2), \tag{119}$$
and
$$\frac{1}{m}\log\det\left(I - \hat{Q}\right) \to n\log A + \log B_0 - \frac{(n+1)\hat{q}_0}{B_0} \quad \text{as } m \to 0, \tag{120}$$
where $B_0 = B|_{m=0} = 1 - \hat{q}_d - n\hat{q}_1 + (n+1)\hat{q}_0$. Combined with $\frac{1}{m}\operatorname{Tr}\left(Q\hat{Q}\right) \to (n+1)\hat{q}_d + n(n+1)q_1\hat{q}_1 - (n+1)^2 q_0\hat{q}_0$, this gives Equation 32.

APPENDIX C The $n \to 0$ expansion of the prior term

We differentiate Equation 32 in n at fixed parameters. Set $L(n) = n\log A + \log B_0 - (n+1)\frac{\hat{q}_0}{B_0}$ and $T(n) = (n+1)\hat{q}_d + n(n+1)q_1\hat{q}_1 - (n+1)^2 q_0\hat{q}_0$. Then
$$\partial_n B_0 = -\hat{q}_1 + \hat{q}_0, \tag{121}$$
$$\partial_n[(n+1)\hat{q}_0/B_0] = \frac{\hat{q}_0}{B_0} - \frac{(n+1)\hat{q}_0(\hat{q}_0 - \hat{q}_1)}{B_0^2}. \tag{122}$$
At $n = 0$, $B_0 \to B := 1 - \hat{q}_d + \hat{q}_0$, hence
$$\partial_n L\mid_{n=0} = \log A - \frac{\hat{q}_1}{B} + \frac{\hat{q}_0(\hat{q}_0 - \hat{q}_1)}{B^2}. \tag{123}$$
Likewise $\partial_n T|_{n=0} = \hat{q}_d + q_1\hat{q}_1 - 2q_0\hat{q}_0$. Combining yields Equation 39.

APPENDIX D Order-n saddle-point equations and Nishimori collapse

The free entropy φ_n vanishes at $n = 0$ when evaluated at the saddle (since $\log Z_0(F) = 0$). Consequently the bare stationarity of $\varphi_n|_{n=0}$ in $(q_0, q_1, \hat{q}_d, \hat{q}_0, \hat{q}_1)$ is degenerate, and the physical equations are extracted from the linear-in-n piece.

The cleanest route — and the one we follow — is to **not** introduce a separate scaling ansatz at the level of Equation 32, but rather to work directly with the finite-n Rényi saddle of Equation 74. That object is by construction evaluated **at** the matrix-inversion identities Equation 71, so the conjugate parameters are no longer independent unknowns and the only scaling assumption we need is on the surviving overlaps (q_0, q_1).

D.1 Bayes-optimal scaling

For the **Bayes-optimal entropy** $S = \mathbb{E}_F H_F(X)$ — matched-prior teacher-student inference — Nishimori symmetry forces $q_0^* = 0$ at $n = 0$ (cross-block independence; see Section 2.5.1) and leaves a single surviving overlap $q_1^* = q^* > 0$. The correct scaling in the $n \to 0$ Shannon limit is therefore

$$q_1 = q + O(n), \quad \hat{q}_1 = \hat{q} + O(n), \quad q_0, \hat{q}_0, \hat{q}_d = O(n), \tag{124}$$

where q and $\hat{q}$ denote the limiting Bayes-optimal values. The earlier ansatz $q_1 = q_0 + n\, r_1 + \ldots$ (with q_0 as the leading scale) is **incompatible** with $q_1^* = q^* > 0$ and is discarded. The matrix-inversion identity Equation 71 at $n = 0$ then forces $\hat{q}_d^* \to 0$ as well: from $1 - q = 1/A$ with $A = 1 - \hat{q}_d + \hat{q}_1$,

$$1 - \hat{q}_d^* + \hat{q}^* = 1/(1-q) = 1 + \hat{q}^* \text{ (using } q = \hat{q}/(1+\hat{q}) \text{ below)} \quad \Longrightarrow \quad \hat{q}_d^* = 0, \tag{125}$$

consistent with Equation 124.

D.2 Order-n expansion of the prior saddle

Inserting Equation 124 into Equation 74 and expanding at order n, the $O(1)$ piece vanishes (as it must, since $S^{\mathrm{RS}}_{\mathrm{prior}}/m\,|_{n=0, m\to 0} = 0$ at the saddle) and the surviving order-n contribution is

$$\mathcal{S}^{\mathrm{RS}}_{\mathrm{prior}}/m\,\Big|^{*}_{m\to 0} = n\, g_{\mathrm{prior}}(q, \hat{q}) + O(n^2), \quad g_{\mathrm{prior}}(q, \hat{q}) := -\frac{1}{2}\, q\hat{q} + \frac{1}{2}\,\hat{q} - \frac{1}{2}\log(1+\hat{q}). \tag{126}$$

The $q_0, \hat{q}_0, \hat{q}_d$ contributions enter only at $O(n^2)$ — their leading $O(n)$ pieces cancel pairwise inside Equation 74 — so the surviving Bayes-optimal action depends only on the two scalars $(q, \hat{q})$.

D.3 Stationarity in $(q, \hat{q})$

The order-n piece of φ_n at the Bayes-optimal saddle is

$$\partial_n\, \varphi_n\,\Big|^{\text{Bayes-opt.}}_{n=0} = g_{\mathrm{prior}}(q, \hat{q}) + \gamma\, \partial_n\, \Phi_n\,\Big|_{n=0} = g_{\mathrm{prior}}(q, \hat{q}) - \gamma\, F(q), \tag{127}$$

with

$$F(q) := \mathbb{E}_{z\sim\mathcal{N}(0,q)}\left[H\left(\tilde{\mathcal{P}}(\cdot\,|z)\right)\right] \tag{128}$$

the conditional output entropy at common-cause variance q (cf. Equation 38). Stationarity gives:

- **In $\hat{q}$** (only g_{prior} depends on $\hat{q}$):

$$\partial_{\hat{q}} g_{\mathrm{prior}} = -\frac{q}{2} + \frac{1}{2} - \frac{1}{2(1+\hat{q})} = 0 \quad \Longrightarrow \quad q = \frac{\hat{q}}{1+\hat{q}}. \tag{129}$$

- **In q:** using $\partial_q g_{\text{prior}} = -\hat{q}/2$ and the heat-equation identity $F'(q) = -(1/2)\,\mathcal{I}(q)$ (see below),

$$-\frac{\hat{q}}{2} - \gamma\, F'(q) = -\frac{\hat{q}}{2} + \frac{\gamma}{2}\,\mathcal{I}(q) = 0 \quad \Longrightarrow \quad \hat{q} = \gamma\,\mathcal{I}(q). \tag{130}$$

Together these reproduce Equation 52.

D.4 Heat-equation derivation of $F'(q)$

Both the Gaussian density $\varphi_q(z) = (2\pi q)^{-1/2}\exp(-z^2/(2q))$ and the noise-augmented channel $\tilde{\mathcal{P}}(x|z) = \int Dw\, P_{\text{out}}(x \mid z + \sqrt{1-q}\,w)$ satisfy heat equations in q:

$$\partial_q \varphi_q(z) = \frac{1}{2}\partial_z^2 \varphi_q(z), \quad \partial_q \tilde{\mathcal{P}}(x|z) = -\frac{1}{2}\partial_z^2 \tilde{\mathcal{P}}(x|z). \tag{131}$$

The opposite signs encode the fact that increasing q broadens the prior and sharpens the channel (the residual variance is $1-q$). Differentiating $F(q) = \int \mathrm{d}z\, \varphi_q(z) \sum_x \left[-\tilde{\mathcal{P}}(x|z)\log\tilde{\mathcal{P}}(x|z)\right]$ in q, integrating by parts twice in z on the first contribution, and using $\sum_x \partial_z^2 \tilde{\mathcal{P}} = \partial_z^2 \sum_x \tilde{\mathcal{P}} = 0$, all "log" terms cancel and one is left with

$$F'(q) = -\frac{1}{2}\int \mathrm{d}z\, \varphi_q(z) \sum_x \frac{\left(\partial_z \tilde{\mathcal{P}}(x|z)\right)^2}{\tilde{\mathcal{P}}}(x|z) = -\frac{1}{2}\,\mathcal{I}(q), \tag{132}$$

with $\mathcal{I}(q)$ the Fisher information of Equation 53. The minus sign is the source of the **positive** coupling $\hat{q} = +\gamma\mathcal{I}(q)$ in the saddle: as q increases the channel becomes sharper, the conditional entropy $F(q)$ decreases, and the missing entropy is transferred to the prior.

D.5 Cross-block Nishimori identities

In the present RS parametrization the Bayes-optimal Shannon limit gives

$$q_0^* = 0, \quad \hat{q}_0^* = 0, \quad \hat{q}_d^* = 0, \quad q_1^* = q^*, \quad \hat{q}_1^* = \hat{q}^*. \tag{133}$$

The vanishing of q_0^* is an exact symmetry property of the Bayes-optimal teacher-student model (one-block marginal collapses to the bare prior at $n = 0$; cf. §2.5.1). The vanishing of the conjugates $\hat{q}_0^*$ and $\hat{q}_d^*$ is then a consequence of the matrix-inversion identities Equation 71 at $n = 0$ together with the scaling Equation 124, **not** of an identity of the form $\hat{q}_0^* = \hat{q}_1^*$.

The latter would arise in a different parametrization that introduces the teacher–student overlap m and the student–student overlap q as separate variables, in which case the standard Nishimori identity reads $\hat{m} = \hat{q}$. Here teacher and student replicas have already been symmetrized inside the $(n+1)$-block — there is no separate m-parameter — so $\hat{m} = \hat{q}$ does not appear in the present saddle equations. The n-expansion uses the collapse above to reduce the five-dimensional system Equation 71 and Equation 73 to the two-dimensional Bayes-optimal saddle Equation 52.

The Fisher information $\mathcal{I}(q)$ of Equation 53 is the natural "signal-to-noise" measure controlling how much information X carries about the latent code Z at fixed overlap q. This is the natural "signal-to-noise" measure controlling how much information X carries about the latent code Z at fixed overlap q.

APPENDIX E One-step replica-symmetry breaking

The action Equation 17 carries two replica indices: the **internal** block index $a = 0, \ldots, n$ (the "molecule" associated with one copy of Z_n) and the **outer** disorder index $\alpha = 1, \ldots, m$ (used to compute $\mathbb{E}_F \log Z_n$). A Parisi instability of the quenched free entropy should first be looked for in the outer α-direction: the genuine Parisi level for $\mathbb{E}_F \log Z_n$ is the one that hierarchically partitions the α-replicas. Internal breaking of the $(n+1)$-block is in principle possible at finite Rényi order, but it is a secondary check, not the primary RSB of the quenched problem. The appendix below sets up the **outer-1RSB** ansatz, derives its free entropy, comments on the Bayes-optimal/symmetric-channel reductions, and applies the framework to the noiseless sign channel of Section 4.

E.1 The outer-1RSB ansatz

We keep the inner $(n+1)$-block replica-symmetric, with within-block off-diagonal overlap q_{in}, and partition the outer index $\alpha = 1, ..., m$ into Parisi groups of size μ. Replicas in the same outer group share the cross-block overlap q_{s}; replicas in different outer groups share the cross-group overlap q_0. Explicitly, with (α, a) a generic replica index and $g(\alpha)$ the Parisi group containing α,

$$Q_{(\alpha,a),(\beta,b)} = \begin{cases} 1 & \alpha = \beta, a = b \\ q_{\text{in}} & \alpha = \beta, a \neq b \\ q_{\text{s}} & \alpha \neq \beta, g(\alpha) = g(\beta) \\ q_0 & \alpha \neq \beta, g(\alpha) \neq g(\beta) \end{cases} \tag{134}$$

with the analogous structure on $\hat{Q}$ in terms of $(\hat{q}_d, \hat{q}_{\text{in}}, \hat{q}_{\text{s}}, \hat{q}_0)$. The RS ansatz of §2.3 is recovered when $q_{\text{s}} = q_0$ (no outer breaking) or in the trivial Parisi limits $\mu \in \{1, m\}$. A genuine outer-RSB instability corresponds to

$$q_{\text{s}} > q_0, \tag{135}$$

not to breaking the $(n+1)$-replica molecule.

E.2 Hierarchical Gaussian decomposition

The covariance Equation 134 admits the four-level Gaussian decomposition

$$h^{\alpha a} = \sqrt{q_0}\, u + \sqrt{q_{\text{s}} - q_0}\, v_{g(\alpha)} + \sqrt{q_{\text{in}} - q_{\text{s}}}\, r^\alpha + \sqrt{1 - q_{\text{in}}}\, w^{\alpha a}, \tag{136}$$

with $u, \{v_g\}, \{r^\alpha\}, \{w^{\alpha a}\}$ mutually independent standard Gaussians at successive levels of the hierarchy: u is shared across all replicas, v_g across all α in the same outer Parisi group, r^α across all a in a single $(n+1)$-block, and $w^{\alpha a}$ is independent at every leaf. The new layer relative to Equation 25 is v_g.

Define the noise-augmented channel

$$\tilde{P}_{q_{\text{in}}}(x \mid z) := \int Dw\, P_{\text{out}}\Big(x \;\Big|\; z + \sqrt{1 - q_{\text{in}}}\, w\Big), \tag{137}$$

and the **block kernel**

$$L_n(u, v) := \int Dr \int \mathrm{d}x\, \tilde{P}_{q_{\text{in}}}\big(x \mid \sqrt{q_0}u + \sqrt{q_{\text{s}} - q_0}v + \sqrt{q_{\text{in}} - q_{\text{s}}}r\big)^{n+1}. \tag{138}$$

The r-integral binds the $(n+1)$ inner replicas of one disorder block into a single observation x; the resulting L_n depends on the outer common-causes (u, v) that are seen by the entire block.

E.3 Output free entropy

Outer 1RSB couples disorder blocks within the same Parisi group through the shared field v_g. Carrying out the integrations layer by layer (first w inside each block, then x to assemble L_n, then v_g across the μ blocks of one Parisi group, then u across all Parisi groups), the disorder-averaged output partition function takes the nested form

$$Z_{\text{out}}^{\text{out-1RSB}} = \int Du \left[\int Dv\, L_n(u, v)^\mu \right]^{m/\mu}. \tag{139}$$

After the standard $m \to 0$ limit, the outer-1RSB output free entropy is

$$\boxed{\Phi^{\text{out-1RSB}}_{n,\mu}(q_0, q_\text{s}, q_\text{in}) = \frac{1}{\mu} \int Du \log \int Dv \, L_n(u,v)^\mu. \qquad (140)}$$

For $q_\text{s} = q_0$ the outer field v drops out of L_n, the inner Dv-integral is trivial, and Equation 140 reduces to the RS kernel Equation 28 (with $q_1 = q_\text{in}$). The trivial Parisi limits $\mu \to 1$ and $\mu \to 0$ likewise recover the RS expression.

The prior contribution comes from diagonalizing $\hat{Q}$ in its outer-1RSB invariant subspaces. The structure parallels Section APPENDIX A but with **four** eigenvalue sectors — the "within outer Parisi group, between $(n+1)$-blocks" sector being the new layer — and yields a four-parameter prior action $\mathcal{S}^{\text{out-1RSB}}_{\text{prior}}(q_0, q_\text{s}, q_\text{in}, \hat{q}_d, \hat{q}_0, \hat{q}_\text{s}, \hat{q}_\text{in})$, extremized together with Equation 140 over the seven scalars and the Parisi block size μ. Stationarity in μ — the Parisi equation $\partial_\mu \varphi^{\text{out-1RSB}}_{n,\mu} = 0$ — is the qualitatively new feature.

E.4 Bayes-optimal Shannon limit

In the Shannon limit $n \to 0$, outer RS is automatic at the order needed for the entropy: $Z_0(F) = 1$ pointwise, so the disorder replicas decouple at order n (cf. §2.5.4) and outer-RSB corrections to the Shannon free entropy enter only at $O(n^2)$. Concretely, $L_0(u,v) = 1$ identically and any μ-dependence in Equation 140 cancels at $O(n)$, so the leading Shannon piece of $\Phi^{\text{out-1RSB}}_{n,\mu}$ does not see the outer Parisi exponent μ at all. The Shannon entropy of Equation 55 is therefore **unaffected** by outer 1RSB at the level of the quenched derivative.

Inner replica symmetry inside the $(n+1)$-block, on the other hand, is a Bayes-optimal ansatz: in the matched teacher-student model the Nishimori identity makes teacher and posterior samples exchangeable, equating the teacher–student and student–student overlap moments and supporting a single inner overlap $q_\text{in} = q^*$ at the saddle. This is the Bayes-optimal RS ansatz, not a theorem: exchangeability does not by itself prove RS — Parisi RSB ensembles are still exchangeable after averaging over the random hierarchy — so the RS formula should be read as exact under the standard hypothesis that the Nishimori saddle is locally stable against replicon and inner-1RSB perturbations. For the dense Gaussian HMM with simple log-concave or convex channels this is the expected situation; for more structured or hard channels the corresponding stability check should be performed.

E.5 Finite Rényi order and symmetric channels

At finite Rényi order $n \neq 0$ the Bayes-optimal interpretation is no longer available — the measure is tilted by $p_F(x)^n$ — and the outer-1RSB ansatz Equation 140 is genuinely an ansatz to be tested. If the channel has the parity symmetry Equation 49, the one-block marginal Equation 62 is even in z and the cross-group overlap satisfies

$$q_0 = 0 \quad \text{on the RS branch, for every } n. \qquad (141)$$

Importantly, this is **not** sufficient to exclude outer 1RSB at finite n: a parity-symmetric problem can still develop an outer Parisi instability through

$$q_\text{s} > q_0 = 0, \qquad (142)$$

in which case Equation 140 has nontrivial v-dependence even though $q_0 = 0$. Symmetry kills the u-channel of the cross-block coupling but leaves the v-channel inside each Parisi group. A full check at finite Rényi order therefore requires either an explicit outer-replicon computation around the RS saddle, or the direct extremization of Equation 140 at $q_0 = 0$ with q_s free.

E.6 Dynamical 1RSB and the hard phase

Independent of the static structure, the action Equation 140 may admit a **secondary**, non-Nishimori saddle with $q_s > q_0$ and $\mu \in (0,1)$. This dynamical saddle is not the global thermodynamic optimum, but it encodes physical structure of the Bayes-optimal posterior $p(z \mid x, F)$ that is invisible to the static thermodynamic free energy:

1. **Clustering of the posterior.** Above a **dynamical 1RSB threshold** γ_d, channel-dependent, the posterior shatters into exponentially many metastable clusters. A typical cluster has weight $e^{-N\Sigma}$, with the **complexity** $\Sigma \geq 0$ counting clusters per unit N.
2. **Information-computation gap.** Local algorithms (belief propagation, AMP, Glauber dynamics) get trapped inside a single cluster and fail to recover the planted z^* even when the static RS overlap q^* is positive. The **hard phase** is the regime $\gamma_d < \gamma < \gamma_s$ where information-theoretic recovery is possible but no efficient algorithm achieves it; below γ_d the problem is “easy”, above γ_s (“static spinodal”) the planted cluster ceases to dominate and recovery is information-theoretically impossible.
3. **Slow MCMC mixing.** Posterior samplers based on local moves have mixing time exponential in N above γ_d, since transitions between clusters require crossing macroscopic free-energy barriers.

The complexity is extracted from the outer-1RSB action by the standard Monasson construction

$$\Sigma(q_0, q_s) = \partial_\mu \varphi_{n,\mu}^{\text{out-1RSB}} \big|_{n=0,\, \mu\to 0^+}, \tag{143}$$

extremized over the cross-group sector at fixed inner overlap $q_{\text{in}} = q^*$ from the RS solution. The dynamical transition γ_d is the smallest γ at which $\Sigma \geq 0$ admits a nontrivial root with $q_s > q_0$. For richer settings — committee machines, sparse priors, low-rank matrix estimation — dynamical 1RSB is well-documented and underlies the algorithmic phase diagrams of the inference literature.

E.7 Application to the noiseless sign channel

We now specialize the framework to the noiseless sign channel of Section 4. The noise-augmented kernel Equation 137 is the probit with variance $1 - q_{\text{in}}$, $\tilde{P}_{q_{\text{in}}}(x \mid z) = \Phi\big(xz/\sqrt{1-q_{\text{in}}}\big)$. With parity symmetry, $q_0 = 0$ on the RS branch. Setting $q_0 = 0$, the block kernel Equation 138 evaluates to

$$L_n(u,v)\,|_{q_0=0} = \sum_{x=\pm 1} \int Dr\, \Phi(x\,\xi)^{n+1}, \quad \xi := \big(\sqrt{q_s}\, v + \sqrt{q_{\text{in}} - q_s}\, r\big)/\sqrt{1-q_{\text{in}}}, \tag{144}$$

and the outer-1RSB output free entropy reduces to a u-independent integral,

$$\Phi_{n,\mu}^{\text{out-1RSB}}\,|_{q_0=0} = \frac{1}{\mu} \log \int Dv \left(\sum_{x=\pm 1} \int Dr\, \Phi(x\,\xi)^{n+1} \right)^{\mu}. \tag{145}$$

E.7.1 Reduction at the Bayes-optimal Shannon point

In the Shannon limit $n \to 0$ with $q_{\text{in}} = q^*$, the inner r-integral and the x-sum recombine because $L_0 = 1$ pointwise. Expanding to first order in n at fixed (q_s, μ), the dependence on μ and q_s in Equation 140 cancels at $O(n)$ — consistent with the general Shannon argument of §2.5.4 — and the conditional-entropy formula Equation 93 is recovered:

$$\partial_n\, \Phi_{n,\mu}^{\text{out-1RSB}}\,|_{n=0,\, q_{\text{in}}=q^*} = -\mathbb{E}_{t\sim\mathcal{N}(0,\, q^*/(1-q^*))}[h_2(\Phi(t))]. \tag{146}$$

At Shannon order, outer 1RSB therefore adds **no information** on top of the RS formula Equation 98, regardless of the value of (q_{s}, μ). This does not, however, prove absence of dynamical 1RSB: the latter is a property of subleading saddles in the action, not of the leading Shannon derivative.

E.7.2 Remarks on stability at finite Rényi order

For the noiseless sign channel the posterior $p(z \mid x, F) \propto p(z) \prod_i \Theta(x_i (Fz)_i)$ is a Gaussian measure restricted to an intersection of half-spaces — a log-concave measure on a convex set. Standard log-concavity arguments make full RS plausible at all $\gamma > 0$ in the Shannon problem, in agreement with the empirical performance of AMP/belief-propagation attaining q^*. At finite Rényi order, however, the tilted measure $p_F(x)^{n+1}$ is not the original posterior, and absence of outer 1RSB requires a separate replicon/stability computation around the $q_0 = 0$ RS saddle — symmetry alone forces $q_0 = 0$ but leaves q_{s} to be checked. We do not perform this stability analysis here; it is left as a natural follow-up to validate the RS Rényi profile of Figure 1 beyond a neighbourhood of $\alpha = 1$.